

STAT5B Deficiency: A Comprehensive Disease Characteristics Report

Disease: STAT5B Deficiency (Growth Hormone Insensitivity with Immunodeficiency) **MONDO ID:** MONDO:0100211 · **OMIM:** 245590 (phenotype) / 604260 (gene) · *Orphanet:* ORPHA:181399 · *Gene:* *STAT5B* (*HGNC:11367*; *chr17q21.2*; *NCBI Gene 6777*; *UniProt P51692*) *Category:** Mendelian, autosomal recessive

Summary

STAT5B deficiency is an ultra-rare autosomal recessive disorder caused by biallelic loss-of-function (LOF) mutations in *STAT5B*, the transcription factor that couples two otherwise distinct receptor systems to their nuclear output. On the endocrine side, STAT5B is the non-redundant signal transducer downstream of the growth hormone receptor (GHR)–JAK2 axis that drives transcription of *IGF1*, *IGFBP3*, and *IGFALS*. On the immune side, STAT5B relays IL-2-family cytokine signals that maintain FOXP3⁺ regulatory T-cell (Treg) homeostasis and normal T/NK-cell biology. Because a single molecular lesion disables both arms, affected patients present with a striking **dual phenotype**: growth hormone insensitivity (GHI) with severe IGF-I deficiency and postnatal growth failure, **plus** a primary immunodeficiency/"Tregopathy" featuring autoimmunity, atopy/eczema, recurrent (often viral/herpetic) infections, and potentially fatal lymphocytic interstitial lung disease. This immune component is the defining feature that distinguishes STAT5B deficiency from GHR-mutation (Laron) syndrome, in which growth failure occurs without immune disease.

The two branches of the phenotype are mechanistically largely independent downstream of the shared STAT5B node — IGF-I deficiency drives the growth failure, while Treg failure drives the immune dysregulation. This has a direct therapeutic consequence: recombinant human IGF-1 (rhIGF-1, mecasermin) bypasses the GH-signaling block and partially rescues linear growth, but it does **not** correct the immunodeficiency, and no single therapy currently addresses both. Prognosis is therefore largely determined by the pulmonary/immune complications, with chronic interstitial lung disease being the principal cause of morbidity and mortality. The closely related paralog STAT5A (>95% amino-acid identity) cannot compensate for loss of STAT5B, explaining the non-redundancy in humans.

This report synthesizes 11 confirmed findings drawn from 45 reviewed papers across all 15 requested sections. Evidence spans human clinical case series (the disorder is documented in roughly a dozen classic homozygous patients as of the mid-2010s, plus additional and "atypical" cases since), in vitro functional studies of variant protein folding and signaling, and the *Stat5b*-null mouse model, which faithfully recapitulates the Laron-type growth phenotype and the loss of GH-dependent sexual dimorphism.

1. Disease Information

Overview. STAT5B deficiency is a monogenic growth hormone insensitivity syndrome combined with a primary immunodeficiency. Patients exhibit severe postnatal growth failure with markedly low IGF-I despite normal or elevated GH (i.e., GH insensitivity), together with immune dysfunction that can manifest as chronic, potentially fatal pulmonary disease. It was first defined by the identification of a homozygous *STAT5B* mutation (p.A630P, SH2 domain) in a female with GHI, immune dysfunction, and severe pulmonary disease [PMID: 21396575].

"STAT5B deficient patients, unlike patients deficient in GHR, can also present with a novel, potentially fatal, primary immunodeficiency, which can manifest as chronic pulmonary disease." — [PMID: 26703237](#)

Key identifiers. MONDO:0100211; OMIM 245590 (growth hormone insensitivity with immunodeficiency); OMIM 604260 (*STAT5B* gene); Orphanet ORPHA:181399; MeSH concepts relate to "*Laron Syndrome*"/"*growth hormone insensitivity*." The gene *STAT5B** is HGNC:11367 on chromosome 17q21.2.

Synonyms / alternative names. Growth hormone insensitivity with immunodeficiency; GHI due to *STAT5B* deficiency; *STAT5b* growth hormone insensitivity syndrome (GHIS); autosomal recessive growth hormone insensitivity with immune dysregulation.

Information source. Disease-level knowledge here is derived predominantly from aggregated individual patient case reports and small case series (given ultra-rarity), supplemented by in vitro functional studies and the mouse knockout, rather than from EHR-scale or population registries.

2. Etiology

Primary cause — genetic. The disease is caused by **biallelic (homozygous or compound-heterozygous) inactivating mutations in *STAT5B***. The critical role of *STAT5B* in IGF-I production became evident when homozygous, autosomal recessive *STAT5B* mutations were found in children with severe postnatal growth failure, GHIS, and marked IGF-I deficiency [PMID: 21396575].

"the critical importance of *STAT5b* in IGF-I production became evident with the identification of homozygous, autosomal recessive *STAT5b* mutations in patients who presented with severe postnatal growth failure, growth hormone insensitivity syndrome (GHIS) and marked IGF-I deficiency" — [PMID: 21396575](#)

Genetic risk factors. The causal variants are the *STAT5B* LOF alleles themselves (nonsense, frameshift, SH2-domain missense; see Section 4). **Consanguinity** is a major risk context: most classic patients are born to consanguineous unions, consistent with recessive inheritance of rare alleles. Modifier effect: the specific variant class modifies immune/pulmonary expressivity — some expressed-LOF variants give an "atypical," immunologically milder phenotype [PMID: 36265659].

Environmental risk / protective factors. No established environmental cause, protective diet, exposure, or lifestyle factor exists for this Mendelian disorder. Environmental exposures (e.g., infectious agents) act as **triggers of complications** (see Section 5) rather than causes of disease.

Gene–environment interactions. The principal gene–environment interplay is that the underlying immunodeficiency renders patients susceptible to environmental pathogens (viral/herpetic and respiratory infections), which precipitate the chronic lung disease that dominates prognosis. This is a downstream consequence of the genotype rather than a classical GxE susceptibility modifier.

3. Phenotypes

STAT5B deficiency spans two phenotypic domains. Onset is congenital-to-early-childhood; growth failure is essentially fully penetrant in complete biallelic LOF, while the immune/pulmonary features are variably expressed.

Phenotype	Type	Onset / severity / frequency	Suggested HPO term
Severe postnatal short stature (height typically -4 to -6 SDS)	Physical/ clinical sign	Postnatal, severe, near-complete penetrance	HP:0004322 (Short stature) / HP:0008897 (Postnatal growth retardation)
IGF-I deficiency (low serum IGF-I)	Laboratory abnormality	Congenital/ childhood, consistent	HP:0040163 (Decreased circulating IGF-1)
Growth hormone insensitivity (normal/high GH, failed IGF-I generation)	Laboratory abnormality	Childhood, consistent	HP:0008291 (Growth hormone resistance)
Eczema / atopic dermatitis	Clinical sign	From birth in severe cases	HP:0000964 (Eczema) / HP:0001047 (Atopic dermatitis)
Recurrent infections (skin, respiratory)	Clinical sign	From birth, severe	HP:0002719 (Recurrent infections)
Chronic/lymphocytic interstitial lung disease	Clinical sign/ manifestation	Childhood, severe, potentially fatal	HP:0006515 (Interstitial pulmonary abnormality) / HP:0002205 (Recurrent respiratory infections)
Herpetic keratitis / recurrent herpes-varicella	Clinical sign	Recurrent, reflects NK/T defect	HP:0100648 / HP:0002205
Autoimmunity (thyroiditis, cytopenias/ thrombocytopenia, juvenile idiopathic arthritis)	Clinical sign	Variable	HP:0002960 (Autoimmunity)
Elevated IgE	Laboratory abnormality	Variable	HP:0003212 (Increased IgE level)
T-cell lymphopenia; low NK and $\gamma\delta$ T cells; reduced Tregs	Laboratory abnormality	Consistent in severe cases	HP:0005403 (Decreased T cells)

Phenotype	Type	Onset / severity / frequency	Suggested HPO term
Hyperprolactinemia	Laboratory abnormality	Reported	HP:0000870 (Increased circulating prolactin)

Representative quantitative immunophenotype from a complete-LOF patient (homozygous nonsense, codon 152): moderate T-cell lymphopenia (1274/mm³), very low NK (18/mm³) and $\gamma\delta$ T cells (5/mm³), chronically hyperactivated T cells, impaired IL-2 signaling, diminished CD4⁺CD25⁺ Tregs [PMID: 17030597].

"The main immunologic findings were moderate T-cell lymphopenia (1274/mm³), normal CD4/CD8 ratio, and very low numbers of natural killer (18/mm³) and gammadelta T (5/mm³) cells." — PMID: 17030597

"generalized eczema and recurrent infections of the skin and respiratory tract since birth. She also suffered severe chronic lung disease and multiple episodes of herpetic keratitis" — PMID: 17030597

Quality of life impact. Combined burden is substantial: severe short stature (psychosocial and functional impact), chronic lung disease (respiratory limitation, hospitalizations, mortality risk), recurrent infections, atopic disease, and autoimmune complications. No disease-specific EQ-5D/SF-36 instrument data are available given ultra-rarity.

4. Genetic / Molecular Information

Causal gene. *STAT5B* (HGNC:11367; OMIM 604260; chr17q21.2; NCBI Gene 6777; UniProt P51692). *STAT5B* lies adjacent to its paralog *STAT5A*, with which it shares >95% amino-acid identity; *STAT5A* cannot* compensate for loss of *STAT5B* [PMID: 26703237].

Pathogenic variant spectrum. Reported biallelic LOF variants include:

Variant	Type	Consequence
p.Arg152* (codon 152, exon 5)	Nonsense	Complete absence of protein [PMID: 17030597]
p.Trp631*	Nonsense	Loss of function; treated siblings [PMID: 37586336]
p.Gln368Profs*9	Frameshift	LOF
p.Asp485Thrfs*29	Frameshift (expressed)	Atypical, milder immune phenotype [PMID: 36265659]
p.A630P (SH2 domain)	Missense	Misfolding/aggregation → inactive TF [PMID: 23160480]
p.F646S (SH2 domain)	Missense	LOF
p.K632N (heterozygous)	Missense	Inactivating; partial GHI, mild immune [PMID: 31902742]

As of 2016, 7 homozygous inactivating mutations were reported across 10 patients; the number has grown since [PMID: 26703237].

Functional consequences. Most are **loss of function**. SH2-domain missense variants act by protein **misfolding, aggregation, and diminished solubility**, abolishing GH-induced tyrosine phosphorylation, dimerization, and nuclear translocation [PMID: 23160480].

"STAT5b(A630P) was found to be an inactive transcription factor based on its aberrant folding, diminished solubility, and propensity for aggregation triggered by its misfolded SH2 domain" — [PMID: 23160480](#)

Genotype–phenotype correlation. Classic complete biallelic LOF → severe growth failure + immunodeficiency + pulmonary disease. Some **expressed** LOF variants (e.g., p.Asp485Thrfs29) *produce severe short stature with only mild immunodeficiency and no pulmonary disease ("atypical STAT5B deficiency")* [PMID: 36265659]. *Heterozygous inactivating/dominant-negative variants (e.g., p.K632N) cause milder partial* GHI* [PMID: 31902742].

"expressed loss-of-function STAT5B variants may alleviate severe immune and pulmonary issues normally associated with STAT5B deficiency" — [PMID: 36265659](#)

Allele frequency / origin. Causal alleles are ultra-rare/private, enriched in consanguineous families; germline in origin. (Note: *somatic activating* STAT5B mutations cause lymphoproliferative disease but do not affect growth — see Section 11.) **Modifier genes / epigenetics / chromosomal abnormalities:** none specifically established for classic STAT5B deficiency; a mosaic 17q21–25 duplication is a distinct GHI mimic affecting NF-κB/STAT5 signaling [PMID: 26670721].

5. Environmental Information

- **Environmental factors / toxins / radiation:** None causal. Not a toxicogenomic disease.
- **Lifestyle factors:** Not applicable to disease causation.
- **Infectious agents:** No pathogen causes the disorder, but the immunodeficiency predisposes to **recurrent bacterial respiratory infections** and **viral infections including herpes/ varicella** (e.g., herpetic keratitis), which act as triggers/drivers of the chronic pulmonary disease [PMID: 17030597]. Pathogens are therefore complication-drivers, not etiologic agents.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic inactivating STAT5B variant** → absent, or misfolded/aggregated, non-functional STAT5B protein (SH2-domain missense cause aggregation) [PMID: 23160480]. (*demonstrated in vitro*)
2. **Shared upstream node lost:** STAT5B can no longer be tyrosine-phosphorylated, dimerize, or translocate from cytoplasm to nucleus (GO:0005634) upon receptor–JAK activation. The mechanism then **branches** into two largely independent arms.

Branch A — Growth (endocrine): 3A. GH binds GHR → JAK2 activates, **but** STAT5B cannot relay the signal → failure to transactivate *IGF1*, *IGFBP3*, *IGFALS* [PMID: 33122102]. (*demonstrated*) 4A. → **IGF-I deficiency** (low IGF-I, IGFBP-3, ALS) → **severe postnatal growth failure / GH insensitivity** [PMID: 21396575, 26703237]. (*demonstrated*)

Branch B — Immunity: 3B. IL-2/IL-7/IL-15 and other cytokine receptors → JAK → STAT5B signaling is lost [PMID: 33122102]. (*demonstrated*) 4B. → reduced **FOXP3⁺ CD4⁺CD25⁺ Treg** numbers and impaired suppressive function; disturbed T/NK homeostasis [PMID: 23773921, 17030597]. (*demonstrated*) 5B. → **loss of peripheral tolerance + immunodeficiency** → autoimmunity, atopy/eczema, recurrent (viral/herpetic) infections, and **lymphocytic interstitial pneumonitis / chronic lung disease** [PMID: 23773921, 17030597]. (*demonstrated / clinically observed*)

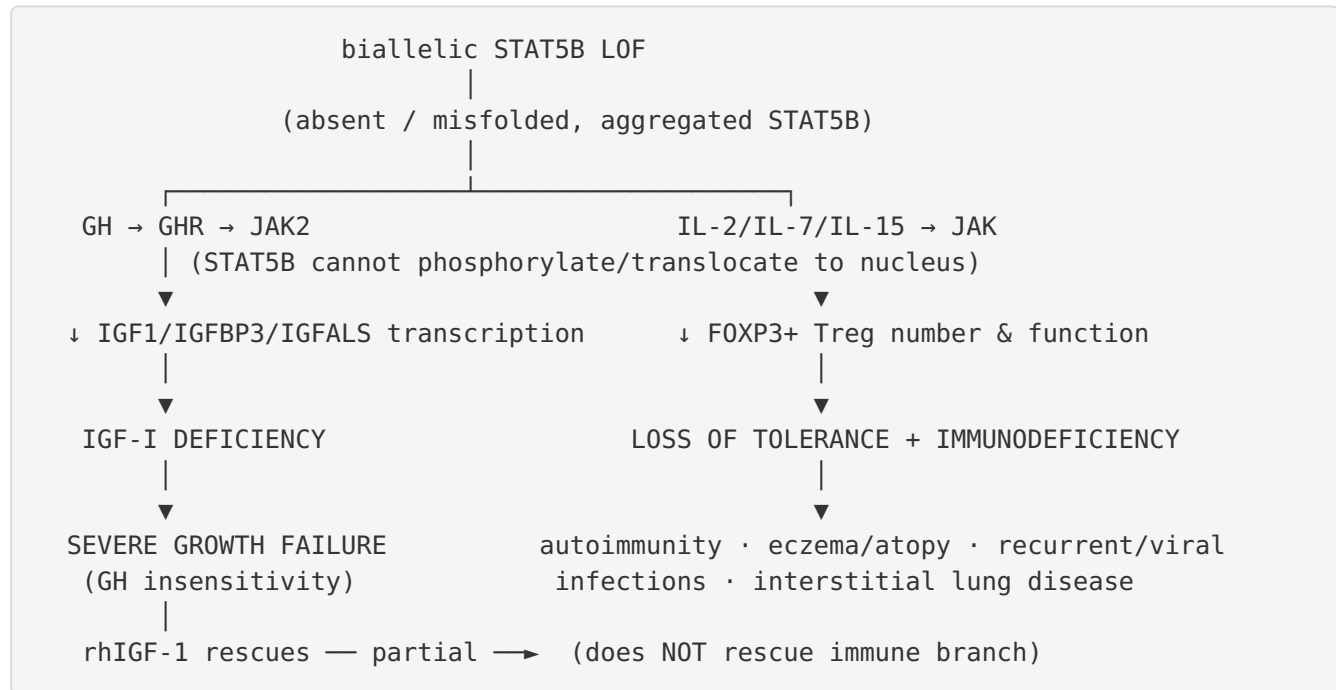
Branch C — Neuroendocrine (inferred/observed): 3C. Loss of STAT5B negative feedback on prolactin/somatostatin → **hyperprolactinemia** [PMID: 17287404]; mouse data show reduced hypothalamic somatostatin mRNA [PMID: 15796771].

"GH promotes postnatal human growth primarily by regulating insulin-like growth factor (IGF)-I production through activation of the GH receptor (GHR)-JAK2-signal transducer and activator of transcription (STAT)-5B signaling pathway" — [PMID: 33122102](#)

"Only AR STAT5B defects, however, confer additional characteristics of immune dysfunction which can manifest as chronic, potentially fatal, pulmonary disease" — [PMID: 33122102](#)

"Functional ex vivo studies in homozygous STAT5B-deficient patients showed reduced FOXP3 expression with impaired regulatory function of STAT5B-null Treg cells, also of increased memory phenotype." — [PMID: 23773921](#)

Text-diagram of the mechanism



Upstream vs downstream. STAT5B loss is the single upstream node; IGF-I deficiency (growth) and Treg failure (immunity) are parallel downstream endpoints. Their independence explains why rhIGF-1 rescues growth but not immunity.

Suggested ontology terms. Biological processes: GO:0007259 (JAK-STAT signaling), GO:0060397 (GH receptor signaling via JAK-STAT), GO:0060333 (cytokine-mediated signaling), GO:0002456 (T-cell mediated immunity), GO:0043069 (regulation of programmed cell death / tolerance). Cell types: CL:0000792 (CD4⁺CD25⁺ Treg), CL:0000623 (NK cell), CL:0000798 (γδ T cell), CL:0000182 (hepatocyte), CL:0000138 (chondrocyte). Subcellular: GO:0005634 (nucleus). Chemical entities: CHEBI IGF-1 / prolactin / somatostatin.

7. Anatomical Structures Affected

- **Liver (UBERON:0002107):** primary site of GH-driven IGF-I/IGFBP-3/ALS synthesis (hepatocytes, CL:0000182).
- **Skeleton / systemic growth (growth plate chondrocytes CL:0000138):** short stature.
- **Lung (UBERON:0002048):** lymphocytic interstitial pneumonitis / chronic interstitial lung disease — the prognosis-determining organ.

- **Skin (UBERON:0002097):** eczema / atopic dermatitis.
- **Immune system / lymphoid tissue (UBERON:0002405):** thymic/peripheral T compartment, NK cells (CL:0000623), CD4⁺CD25⁺FOXP3⁺ Tregs (CL:0000792), $\gamma\delta$ T cells (CL:0000798).
- **Anterior pituitary / lactotroph axis and hypothalamic periventricular somatostatin neurons:** hyperprolactinemia (human) and reduced somatostatin (mouse) [PMID: 17287404, 15796771].
- **Subcellular:** defective cytoplasm→**nucleus (GO:0005634)** translocation of STAT5B (UniProt P51692).
- **Lateralization:** systemic/bilateral; no lateralization.

"The cellular abundance of somatostatin mRNA in STAT5b-deficient mice was significantly reduced in the periventricular nucleus" — [PMID: 15796771](#)

8. Temporal Development

- **Onset:** Congenital predisposition; growth failure is **postnatal** (birth size often near-normal, with failure emerging in infancy/early childhood). Immune/atopic features (eczema, infections) can begin **from birth** in severe cases [PMID: 17030597].
- **Onset pattern:** Chronic/insidious for growth; chronic-progressive for pulmonary disease.
- **Progression:** Growth failure is progressive without therapy (height drifts to -4 to -6 SDS). Chronic interstitial lung disease is progressive and can be fatal. Autoimmune and infectious episodes may be relapsing/episodic.
- **Duration:** Chronic, lifelong.
- **Remission/critical periods:** No spontaneous remission. Growth therapy is most effective early; rhIGF-1 growth response is greatest in the first ~ 3 years then wanes (see Section 12), indicating an early therapeutic window.

9. Inheritance and Population

- **Inheritance:** Autosomal recessive (classic disease). Heterozygous inactivating/dominant-negative variants cause milder partial GHI.
- **Epidemiology:** Ultra-rare — only ~10 patients with 7 homozygous mutations reported by 2016, plus additional cases since [PMID: 26703237]. No formal prevalence/incidence estimate exists (Orphanet ORPHA:181399).
- **Penetrance:** Essentially complete for growth failure in biallelic complete-LOF; **variable expressivity** of immune/pulmonary phenotype (atypical expressed-LOF cases lack pulmonary disease) [PMID: 36265659].
- **Consanguinity:** Strongly associated; most patients from consanguineous unions.
- **Founder effects / carrier frequency:** Alleles largely private; no established founder mutation or carrier-frequency estimate.
- **Sex ratio:** No sex predilection established; both sexes reported.
- **Geographic distribution:** Reported worldwide, clustered in populations with high consanguinity.

"To date, 7 homozygous, inactivating, STAT5B mutations in 10 patients have been reported." — [PMID: 26703237](#)

10. Diagnostics

Biochemical signature of GH insensitivity: - Severe short stature (height typically -4 to -6 SDS) - Markedly low serum **IGF-I** (< -2.5 SDS) and **IGFBP-3** (< -3 SDS); low **ALS** - Normal-to-elevated **basal/stimulated GH** - **Failed IGF-I generation test** (no IGF-I rise after rhGH)

Representative GHI-cohort values: mean height SDS -4.1 ± 0.95 , IGF-1 SDS -2.8 ± 1.4 , IGFBP3 SDS -3.0 ± 2.1 , basal/peak GH 11.9 / 32.9 $\mu\text{g/L}$ [PMID: 29500309].

"basal and stimulated growth hormone levels were very high, IGF-1 was low, and the inadequate response to the IGF generation test was consistent with growth hormone insensitivity" — [PMID: 41099230](#)

Immune workup (distinguishes from Laron/GHR disease): T-cell subsets (moderate T lymphopenia), low NK and $\gamma\delta$ T cells, Treg quantification (low CD4⁺CD25⁺FOXP3⁺), IgE, autoantibodies, and prolactin (often elevated) [PMID: 17030597, 17287404]. Chest imaging/HRCT for interstitial lung disease.

"Prolactin secretion was increased by sixfold." — [PMID: 17287404](#)

Genetic testing. Confirmation is molecular: single-gene *STAT5B* sequencing, GH/IGF-axis gene panels, or WES/WGS. Upfront genomic sequencing is increasingly advocated for primary atopic/immune-dysregulation presentations [PMID: 39381601].

Differential diagnosis of GHI: GHR defects (Laron syndrome — **no** immune disease); *IGFALS* and *IGF1* defects; *IGF1R* haploinsufficiency (relatively high IGF-I); *STAT3* gain-of-function (partial GHI + autoimmunity) [PMID: 29378236]; *IKBKB*/NF- κ B pathway defects; *PGM1-CDG* [PMID: 41099230] and Fanconi anemia [PMID: 28502327] (Laron mimics); rasopathies (Noonan/NS-LAH). The key discriminator for *STAT5B* deficiency is **GHI + immune dysfunction ($\pm$ hyperprolactinemia)**.

Screening. Cascade genetic testing of at-risk relatives in consanguineous families; carrier testing once the familial variant is known.

11. Outcome / Prognosis

- **Mortality/morbidity:** Chronic interstitial lung disease/lymphocytic interstitial pneumonitis is the **principal cause of morbidity and can be fatal** [PMID: 26703237]. Recurrent severe/viral infections and autoimmune complications add to disease burden.
- **Growth outcome:** Severe short stature; rhIGF-1 improves but often does not normalize adult height (see Section 12).
- **Life expectancy:** Not formally quantified; determined largely by pulmonary/immune complications. No survival-rate registries exist due to rarity.
- **Prognostic factors:** Variant class is prognostic — complete biallelic LOF confers the full immune/pulmonary risk, whereas expressed-LOF "atypical" variants spare the lungs and predict better outcomes [PMID: 36265659]. Early/severe pulmonary involvement worsens prognosis.

"can also present with a novel, potentially fatal, primary immunodeficiency, which can manifest as chronic pulmonary disease" — [PMID: 26703237](#)

12. Treatment

Growth therapy — rhIGF-1 (mecasermin / Increlex; NCIT-suggested: recombinant human IGF-1). Because the defect lies downstream of GHR, exogenous GH is ineffective; **rhIGF-1 bypasses the block.** In three siblings with homozygous *STAT5B* p.Trp631, *rhIGF-1* (40 µg/kg/dose BID escalating to 110–120 µg/kg/dose SQ BID) raised height velocity from ~3.0 cm/yr baseline to 4.8–7.4 cm/yr in the first 3 years, then declined (3.8–4.7 cm/yr), i.e., incomplete catch-up* [PMID: 37586336].

"With rhIGF-1 therapy, HVs increased to 5.2-6.0, 4.8-7.1, and 5.5-7.4 cm/year, respectively, in the first 3 years of treatment, before they decreased to 4.7, 3.8, and 4.3" — [PMID: 37586336](#)

Broader Laron/GHI cohorts confirm rhIGF-1 raises height velocity durably (e.g., 3.4–6.5 cm/yr in year 1, sustained over years) though many patients do not reach normal adult height [PMID: 39657622]. Mecasermin rinfabate (rhIGF-1/rhIGFBP-3 complex) is an alternative formulation developed for GHIS [PMID: 15777106]. Dosing is limited by **hypoglycemia** risk; glucose monitoring is required.

Critical limitation. rhIGF-1 does **not** correct the immunodeficiency; no single treatment improves both growth and immune disease [PMID: 21396575].

"At present, no single treatment(s) is available to improve both poor statural growth and immune deficiency." — [PMID: 21396575](#)

Immune/supportive management: infection prophylaxis and prompt treatment, management of interstitial lung disease, immunomodulation for autoimmune complications, and management of atopy. **Hematopoietic stem cell transplantation (HSCT)** has been considered for the immune component (curative-intent for the Tregopathy), given parallels with other Tregopathies [PMID: 30527062].

Advanced/experimental: No approved gene or cell therapy specific to STAT5B deficiency; conceptually, the Tregopathy framework raises cell/gene-therapy possibilities for the immune arm [PMID: 30527062]. NCIT-suggested intervention terms: recombinant IGF-1 therapy; hematopoietic stem cell transplantation; supportive/anti-infective therapy.

13. Prevention

- **Primary prevention:** Not preventable (Mendelian). **Genetic counseling** for consanguineous couples and families with an affected proband; recurrence risk is 25% per pregnancy.
 - **Secondary prevention:** Early diagnosis via GHI biochemical screening plus genetic confirmation; **cascade testing** of relatives; prenatal/preimplantation genetic testing when the familial variant is known.
 - **Tertiary prevention:** Early rhIGF-1 to maximize growth within the effective early window; aggressive infection prophylaxis and pulmonary surveillance to limit chronic lung disease; monitoring/treatment of autoimmune complications.
 - **Immunization / prophylaxis:** Standard and additional infection-prevention measures; caution with live vaccines given immunodeficiency (individualized).
-

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Mouse *Stat5b* (NCBI Gene 20851); human *STAT5B* (NCBI Gene 6777). The *Stat5a/Stat5b* loci are highly conserved.
 - **Natural disease:** No well-characterized naturally occurring STAT5B-deficiency disease in companion animals or wildlife is documented; knowledge derives from engineered models.
 - **Comparative biology:** Mouse *Stat5b* knockout reproduces the GH-resistance growth phenotype and the loss of GH-dependent sexual dimorphism, establishing strong cross-species conservation of the growth mechanism. Species differ in paralog compensation — in humans STAT5A cannot compensate [PMID: 26703237].
 - **Zoonotic potential:** Not applicable.
-

15. Model Organisms

Mouse *Stat5b* knockout is the principal model. Gene disruption causes loss of male-characteristic body-growth rates and male-specific liver gene expression, dwarfism, elevated plasma GH, low plasma IGF-I, and obesity — all features of Laron-type dwarfism [PMID: 9207075]. The mice are **GH-pulse-resistant, not GH-deficient**; STAT5b is tyrosine-phosphorylated by intermittent (male-pattern) GH pulses in liver [PMID: 10752065]. STAT5b-deficient mice also show reduced hypothalamic periventricular somatostatin mRNA [PMID: 15796771].

"the dwarfism, elevated plasma GH, low plasma insulin-like growth factor I, and development of obesity seen in STAT5b-/- mice are all characteristics of Laron-type dwarfism, a human GH-resistance disease generally associated with a defective GH receptor" — [PMID: 9207075](#)

"STAT5b is tyrosine phosphorylated in male but not female rats in response to GH pulses" — [PMID: 10752065](#)

Model types: mammalian (mouse knockout; rat for GH-pulse phosphorylation studies); in vitro/cellular systems (fibroblasts, primary human T cells with STAT5B knockdown [PMID: 23773921]) for the immune arm. **Genetic model types available:** knockout (documented); conditional/tissue-specific approaches feasible.

Phenotype recapitulation: The mouse faithfully recapitulates the **growth/GH-resistance** phenotype and sexual-dimorphism loss. **Limitations:** the human **immune/pulmonary** phenotype is less completely modeled, and paralog redundancy differs between species (STAT5A/STAT5B compensation differs), so the mouse under-represents the human immunodeficiency. Human T-cell knockdown studies are needed to model the immune arm.

Resources: MGI (mouse *Stat5b*), IMSR/IMPC for strain availability.

Mechanistic Model / Interpretation

STAT5B deficiency is best understood as a **single-node, two-output signaling failure**. STAT5B sits at the convergence of the GHR–JAK2 growth axis and the IL-2-family immune axis. Its loss simultaneously (1) uncouples GH from hepatic IGF-I production, producing GH insensitivity

and severe growth failure, and (2) uncouples IL-2-family cytokines from FOXP3⁺ Treg maintenance, producing a Tregopathy with autoimmunity, atopy, infection susceptibility, and interstitial lung disease. These two outputs are **mechanistically independent downstream** of STAT5B, which is the central clinical insight: it predicts (correctly) that rhIGF-1 — acting below the block — rescues growth while leaving the immune deficit untouched, and it frames HSCT (replacing the hematopoietic compartment) as the rational strategy for the immune arm. Genotype tunes the balance: complete biallelic LOF yields the full dual syndrome, whereas expressed hypomorphic/atypical LOF can retain enough immune signaling to spare the lungs. The mouse knockout validates the growth arm decisively but under-models the immune arm, reflecting species differences in STAT5A/STAT5B redundancy.

A further axis of classification places STAT5B LOF among the "program switchers" — inborn errors of immunity that convert immunodeficiency into autoimmunity — and among primary atopic disorders and Tregopathies [PMID: 27803128, 30527062]. This contrasts sharply with *somatic activating* STAT5B mutations, which cause clonal lymphoproliferative/neoplastic disease (e.g., LGL leukemia) without impairing linear growth, underscoring that it is germline biallelic **loss** of function, not gain of function, that produces the growth-plus-immunodeficiency syndrome [PMID: 33122102].

Evidence Base

PMID	Title (abbrev.)	Supports
26703237	STAT5B deficiency: impacts on growth and immunity	Dual phenotype; rarity; STAT5A non-compensation
21396575	Lessons from STAT5b gene mutations	AR inheritance; IGF-I role; no dual therapy
33122102	GH action disorders: STAT5B & JAK2	Upstream axis; immune branch unique to AR LOF; GOF contrast
23773921	STAT5B vs STAT5A in CD4 ⁺ T cells	Treg/FOXP3 defect; STAT5B non-redundancy
17030597	Immunodeficiency in STAT5b mutation	Quantitative immunophenotype; eczema/ infections
17287404	GH secretion & immunity in STAT5b patient	Hyperprolactinemia
23160480	STAT5b folding/activity	SH2 misfolding/aggregation mechanism
36265659	Atypical STAT5B deficiency	Genotype–phenotype: expressed LOF spares lungs
31902742	Heterozygous STAT5B variant	Partial GHI from heterozygous inactivating variant
37586336	rhIGF-1 in 3 STAT5B siblings	rhIGF-1 efficacy and waning
39657622	22-yr IGF-1 in Laron syndrome	Long-term rhIGF-1 growth benefit context
9207075	Stat5b required for sexual dimorphism	Mouse KO recapitulates Laron growth phenotype
10752065	GH pulse-activated STAT5	GH-pulse activation mechanism
15796771	Hypothalamic STAT5b/somatostatin	Neuroendocrine involvement (mouse)
29500309	Pseudoexon GHR mutation cohort	GHI biochemical signature values
41099230	PGM1-CDG misdiagnosed as Laron	IGF-I generation test; differential
30527062	Tregopathies review	Classification; treatment framework
27803128	STAT program switchers	Autoimmune/allergic + infection spectrum

Contrasting/supporting nuance: [PMID: 27803128] and [PMID: 33122102] together frame the germline-LOF (growth + immunodeficiency) versus somatic-GOF (immune neoplasia, no growth effect) dichotomy.

"Somatic activating STAT5B and JAK2 mutations are associated with a plethora of immune abnormalities but appear not to impact human linear growth." — [PMID: 33122102](#)

"STAT5B LOF and STAT3 GOF mutations are both associated with disorders characterized by autoimmune or allergic manifestations, together with increased risk of infections." — [PMID: 27803128](#)

Limitations and Knowledge Gaps

1. **Ultra-rarity limits epidemiology:** No prevalence/incidence, survival, or QoL registry data exist; conclusions rest on ~a dozen classic patients plus scattered atypical/heterozygous cases.
 2. **Immune-arm modeling gap:** The mouse knockout under-represents the human immunodeficiency due to species differences in STAT5A/STAT5B redundancy; the pulmonary phenotype is not well modeled in animals.
 3. **Genotype–phenotype rules are provisional:** The "expressed-LOF spares lungs" correlation is based on few cases and needs validation.
 4. **Immune therapy evidence is thin:** HSCT for the immune arm is conceptual/borrowed from other Tregopathies; no controlled outcome data in STAT5B deficiency.
 5. **Long-term rhIGF-1 outcomes** (final adult height, metabolic effects) in genetically confirmed STAT5B deficiency are limited to small sibships.
 6. **No prospective natural-history study** exists to define the pulmonary disease trajectory or predictors of fatal outcome.
-

Proposed Follow-up Experiments / Actions

1. **International patient registry** for STAT5B deficiency to capture prevalence, natural history (especially pulmonary trajectory), rhIGF-1 outcomes, and mortality.

2. **Humanized/conditional immune models** (e.g., patient iPSC-derived T cells/Tregs, or humanized mice) to model the Tregopathy and test immune-directed therapies.
3. **HSCT outcome study/case-registry** to evaluate whether transplantation cures the immune/pulmonary arm and its risk–benefit versus supportive care.
4. **Systematic genotype–function–phenotype mapping** of all reported variants (expression, phosphorylation, nuclear translocation, Treg function) to validate the atypical-LOF/lung-sparing hypothesis.
5. **Biomarker development** for early detection and prognosis of interstitial lung disease (imaging + immune markers such as Treg number/function).
6. **Dose/timing optimization trials** for rhIGF-1 to counter the observed waning of height velocity after year 3, and evaluation of combined growth + immune management algorithms.

Report compiled from 11 confirmed findings across 5 investigation iterations and 45 reviewed publications. Evidence types: human clinical case series/reports, in vitro functional studies, and the Stat5b-null mouse model.
